Contents

Syllabus

Network Science Module

Lecture 1: Introduction to the course

Overview of the course; introduction to networks

[slides]

reading material: Sec. 0.1-0.6 (Menczer, Fortunato, and Davis 2020) Sec. 1.1, 1.2 (Easley and Kleinberg 2010)

Lectures 2–3: Recap on graphs

Graph theory: basic definitions; density and sparsity; subnetworks; degree; directed networks; weighted networks; multilayer and temporal networks; network representations; paths and distances; BFS; connectedness and components; small world phenomena; clustering coefficient; transitivity

[slides] [notebook]

reading material: Chapter 1–2 (Menczer, Fortunato, and Davis 2020) Sec. 2.1, 2.2, 2.3 (Easley and Kleinberg 2010)

Lecture 4: Strong and weak ties

Strong and weak ties; triadic closure; the strength of weak ties; tie strength and network structure in large-scale data; tie strength, social media, and passive engagement; closure, structural holes, and social capital

[slides] [notebook: neighborhood overlap, local bridges, structural holes]

reading material: Sec. 3.1-3.5 (Easley and Kleinberg 2010)

Lecture 5: Homophily and Structural Balance

Homophily; assortativity; mechanisms underlying homophily: selection and social influence; affiliation; tracking link formation in online data; a spatial model of segregation; signed graphs; structural balance; balance theorem; weak balance

[slides] [notebook: homophily, assortativity, Schelling model]

reading material: Sec. 2.1 (Menczer, Fortunato, and Davis 2020) Sec. 4.1-4.5 (Easley and Kleinberg 2010) Sec. 5.1-5.3 (Easley and Kleinberg 2010)

Lecture 6: Centrality Measures

Hubs; degree centrality; eigenvector centrality; Katz centrality; closeness centrality; betweenness centrality

[slides] [notebook: eigenvector centrality] [notebook: Katz centrality] [notebook: centrality comparison]

reading material: Chapter 3 (Menczer, Fortunato, and Davis 2020)

Lecture 7: Degree Distributions, Scale-Free Networks, and Robustness

Degree distributions; PDF, CDF, CCDF; logarithmic scale; power laws; scale-free networks; heterogeneity parameter; friendship paradox; robustness

[slides] [notebook: degree distributions] [notebook: friendship paradox] [notebook: robustness]

reading material: Chapter 3 (Menczer, Fortunato, and Davis 2020)

Lecture 8: Network Models

Random graphs; Erdős–Rényi / Gilbert model; giant-component phase transition; small-world networks; Watts–Strogatz model; the configuration model; degree-preserving null models

[slides] [notebook: configuration model] [notebook: Erdős–Rényi random graphs] [notebook: Watts–Strogatz]

reading material: Chapter 5 (Menczer, Fortunato, and Davis 2020)

Lecture 9: Preferential Attachment

Network growth; preferential attachment; Polya's urn; Barabási–Albert model; why growth alone is not enough; non-linear preferential attachment; limitations of the BA model

[slides] [notebook: Barabási–Albert model]

Lecture 10: Extensions of Preferential Attachment

Attractiveness model; fitness model; random walk model; copy model; model comparison and selection

[slides] [notebook: attractiveness model] [notebook: fitness model] [notebook: random walk model] [notebook: copy model]

Lecture 11: Power Laws and Rich-Get-Richer Phenomena

Popularity as a network phenomenon; power laws; rich-get-richer models; unpredictability of rich-get-richer effects; the long tail; the effect of search tools and recommendation systems

[slides] [notebook: power law fitting] [notebook: rich-get-richer simulation]

reading material: Chapter 18 (18.1-18.7) (Easley and Kleinberg 2010)

Lecture 12: Communities

Concepts, definitions, and mesoscopic structure; why communities matter; cohesion, connectedness, and separation; strong and weak communities; partitions; Bell numbers; overlapping communities; hierarchical communities; communities versus clustering and graph partitioning; preview of algorithm families

[**slides**]

reading material: Chapter 6 (Menczer, Fortunato, and Davis 2020)

Lecture 13: Community Detection I

Partitioning baselines; minimum cut intuition; Kernighan-Lin; edge betweenness; Girvan-Newman; quality functions; modularity; modularity null model; modularity matrix and spectral ideas; Clauset-Newman-Moore greedy optimization; modularity limits, including random baselines, the resolution limit, and degeneracy

[**slides**] [**notebook: modularity**] [**notebook: Kernighan–Lin**] [**notebook: Girvan–Newman**] [**notebook: Clauset–Newman–Moore**]

reading material: Chapter 6 (Menczer, Fortunato, and Davis 2020)

Lecture 14: Community Detection II

Scalable community detection; Louvain's multilevel modularity optimization; Leiden's refinement and connectivity guarantees; label propagation as local consensus; Infomap and the map equation; comparison between modularity communities, flow communities, and local-consensus communities

[**slides**] [**notebook: Louvain**] [**notebook: label propagation**]

reading material: Chapter 6 (Menczer, Fortunato, and Davis 2020)

Lecture 15: Community Detection III

Statistical inference and validation; stochastic block models; degree-corrected stochastic block models; detectability; planted partition, Girvan-Newman, and LFR benchmarks; real benchmarks and metadata caveats; NMI, AMI, and ARI; negative controls; robustness and reporting practice

[slides] [notebook: stochastic block model] [notebook: community detection benchmark]

reading material: Chapter 6 (Menczer, Fortunato, and Davis 2020)

Lecture 16: Games on Networks and Strategic Interaction

Strategic interaction in complex systems; why game theory belongs in network science; players, strategies, payoffs, and payoff matrices; best responses and dominant strategies; Prisoner's Dilemma; Nash equilibrium; multiple equilibria and coordination games; Hawk-Dove game; mixed strategies; Pareto optimality and social optimality; the gap between local incentives and collective welfare

[slides]

reading material: Chapter 6 (6.1-6.9) (Easley and Kleinberg 2010)

Lecture 17: Traffic Networks and Selfish Routing

Traffic as a complex system; traffic networks as games; roads as edges and routes as strategies; congestion externalities; equilibrium traffic patterns; social cost and social optimum; Braess's paradox; best-response dynamics; potential functions; efficiency loss, price of anarchy versus price of stability, and policy implications

[slides]

reading material: Chapter 8 (8.1-8.3) (Easley and Kleinberg 2010)

Lecture 18: Cascading Behaviors in Networks

Diffusion in networks; modeling diffusion through a network; cascades and clusters; diffusion, thresholds, and the role of weak ties; extensions of the basic cascade model

[slides]

reading material: Chapter 19 (19.1-19.5) (Easley and Kleinberg 2010)

Lecture 19: Link Analysis and Web Search

The structure of the Web; searching the Web; link analysis; HITS: hubs and authorities; PageRank; random walks and PageRank; practical implications; modern Web search; link analysis beyond the Web

[slides]

reading material: Chapter 13 (Easley and Kleinberg 2010) Chapter 14 (14.1-14.5) (Easley and Kleinberg 2010) Chapter 4 (Menczer, Fortunato, and Davis 2020)

Lecture 20: Traditional Machine Learning on Graphs

Traditional ML pipeline on graphs; node-level features: degree, centrality, clustering coefficient, graphlets and GDV; link-level features: distance-based, local overlap (common neighbors, Jaccard, Adamic-Adar, preferential attachment), global overlap (Katz index); graph-level features; kernel methods; graphlet kernel; Weisfeiler-Lehman kernel

[slides] [notebook: traditional ML on graphs (CORA node classification)] [notebook: link prediction (Les Misérables)]

reading material: Jure Leskovec, **CS224W: Machine Learning with Graphs — Traditional Feature-based Methods**

Lecture 21: Representation Learning on Graphs

Graph representation learning; node embeddings; shallow encodings; random-walk embeddings; DeepWalk; node2vec; embedding entire graphs

[slides] [notebook: node2vec]

Lecture 22: Computational Epidemiology — Compartmental Models

Epidemic processes; SI model; SIR model; SIS model; homogeneous mixing; stochastic models

[slides] [notebook: epidemic models]

Lecture 23: Computational Epidemiology — Network Models

Beyond homogeneous mixing; network representation in epidemiology; epidemic spreading on heterogeneous networks; high-resolution contact networks

[slides]

References

Easley, D., and J. Kleinberg. 2010. *Networks, Crowds, and Markets: Reasoning about a Highly Connected World*. Cambridge University Press. https://books.google.it/books?id=atfCl2agdi8C.

Menczer, F., S. Fortunato, and C. A. Davis. 2020. *A First Course in Network Science*. Cambridge University Press. https://books.google.it/books?id=q1abxgEACAAJ.